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### **DIE INTERCONNECT STRUCTURE WITH EMBEDDED INDUCTOR(S) INCLUDING COUPLED COILS FORMED IN REDISTRIBUTION LAYER (RDL) AND ADJACENT BUMP-LEVEL DISTRIBUTION LAYER (BDL) FOR IMPROVED Q FACTOR**

#### **Abstract**

Aspects disclosed herein include a die interconnect structure with embedded inductor(s) including coupled coils formed in redistribution layer (RDL) and adjacent bump-level distribution layer (BDL) for improved Q factor. The die interconnect structure includes a metal pad and a first coil formed in the RDL. The die interconnect structure also includes the BDL which includes a secondary metal pad and a second coil. The die interconnect structure also includes a seed layer between the first coil and the second coil. The die interconnect structure includes an inductor comprising the first coil, the second coil, and a via coupling the first and second coils. In this manner, the additional second coil formed in the BDL has higher conductance, less resistance than a coil formed in metallization layers within the die and, thus, advantageously improving the Q factor of the inductor.

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## Background/Summary

### BACKGROUND

#### I. Field of the Disclosure

[0001] The field of the disclosure relates to integrated circuit (IC) packages that include one or more semiconductor dies (“dies”) attached to a package substrate, and more particularly to inductors formed in a die interconnect structure on a die(s).

#### II. Background

[0002] Integrated circuits (ICs) are the cornerstone of electronic devices. ICs are packaged in an IC package, also called a “semiconductor package” or “chip package.” The IC package includes one or more semiconductor dice (“dies” or “dice”) as an IC(s) that is mounted on and electrically coupled to a package substrate to provide physical support and an electrical interface to the die(s). The package substrate also includes one or more metallization layers that include metal interconnects (e.g., metal traces, metal lines) with vertical interconnect accesses (vias) coupling the metal interconnects together between adjacent metallization layers to provide electrical interfaces between the die(s). The package substrate also includes an outer metallization layer that includes metal interconnects coupled to external metal interconnects (e.g., ball grid array (BGA) interconnects) to provide an external interface between the die(s) in the IC package. The external metal interconnects can also be coupled (e.g., soldered) to traces in a printed circuit board (PCB) to attach the package to the PCB and interface its die(s) with the circuitry of the PCB.

[0003] The die(s) also includes one or more metallization layers that include metal interconnects (e.g., metal traces, metal lines). An outer metallization layer, also known as a redistribution layer (RDL), includes metal interconnects (e.g., metal pads, metal traces, and metal lines). The die(s) also includes die interconnects (e.g., balls or pillars) which electrically couple the metal interconnects in the die(s) to the metal interconnects exposed in the outer metallization layer of the package substrate. Inductors are typically formed in the one or more metallization layers in the die. [0004] A Q factor characterizes the performance of an inductor at a given frequency and is defined as the ratio of the inductive reactance to the series resistance of the inductor. A higher Q factor indicates a lower rate of energy loss and a more efficient performance of the inductor. The Q factor is calculated using the formula  $Q=R\omega/L$ , where  $L$  is the inductance,  $R$  is the DC resistance, and  $\omega L$  is the inductive reactance. Lowering the resistance of the inductor improves its Q factor and narrows its bandwidth, leading to better performance. Unwanted resistance in an inductor can be due to factors such as standard DC resistance, skin effect, radiated energy, and core losses.

Therefore, to improve the Q factor of an inductor on a semiconductor, it is important to minimize these resistance factors and optimize the inductive reactance. However, increasing conductance (i.e.,  $1/R$ ) and thus increasing a Q factor can be challenging in IC package designs given the limited number and widths of metallization layers formed within a die or in the RDL of the die.

### SUMMARY

[0005] Aspects disclosed in the detailed description include a die interconnect structure with embedded inductor(s) including coupled coils formed in a redistribution layer (RDL) and adjacent

bump-level distribution layer (BDL) for improved Q factor. Related integrated circuit (IC) packages and fabrication methods are also disclosed. The die interconnect structure formed on a die includes the RDL which includes a metal pad and a first coil. The RDL could be a back-end-of-line (BEOL) interconnect structure of a semiconductor die as an example. The die interconnect structure also includes the BDL which includes a secondary metal pad and a second coil. The BDL is a metallization layer which is as thick or thicker than the RDL. The die interconnect structure also includes a seed layer between the first coil and the second coil and is used in depositing the BDL during fabrication and, in particular, forming the die interconnect and second coil. The BDL may be manufactured at a semiconductor and test (SAT) facility. The die interconnect structure includes an inductor comprising the first coil, the second coil, and a via coupling the first and second coils. Rather than be limited to the number of metallization layers in the die and their smaller thickness than the BDL, the additional second coil is also formed in the same thick BDL containing the die interconnects and is coupled to the first coil formed in the RDL through a via. In this manner, the additional second coil formed in the thick BDL has higher conductance and less resistance than a coil formed in metallization layers within the die and, thus, advantageously improving the Q factor of the inductor.

[0006] In this regard in one aspect, a die interconnect structure formed on a die comprises a first surface. The die interconnect structure comprises a RDL extending in a first direction adjacent to the first surface and having a second surface. The RDL comprises a first coil extending in the first direction. The die interconnect structure further comprises a BDL extending in the first direction. The BDL comprises a second coil. The die interconnect structure further comprises a seed layer extending in the first direction adjacent to the second surface of the RDL, the seed layer between the first coil and the second coil. The die interconnect structure further comprises an inductor. The inductor comprises the first coil, the second coil, and a via extending in a second direction orthogonal to the first direction, the via coupling the first coil to the second coil.

[0007] In another aspect, a method for forming a die interconnect structure formed on a die comprising a first surface is disclosed. The method comprises depositing a RDL extending in a first direction adjacent to the first surface and having a second surface. Depositing the RDL comprises forming a first coil extending in the first direction. The method further comprises depositing a seed layer extending in the first direction adjacent to the second surface of the RDL and depositing a BDL extending in the first direction. Depositing the BDL further comprises forming a second coil, the seed layer between the first coil and the second coil. The method further comprises forming an inductor. Forming the inductor comprises forming a via extending in a second direction orthogonal to the first direction, the via coupling the first coil to the second coil.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a cross-sectional side view of an exemplary IC package that includes a die interconnect structure with an embedded inductor including coupled coils formed in a redistribution layer (RDL) and adjacent bump-level distribution layer (BDL) for improved Q factor;

[0009] FIG. 2A is a close-up cross-sectional side view of a die shown in FIG. 1 having an exemplary die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor;

[0010] FIG. 2B is a close-up view of another exemplary die interconnect structure with an embedded inductor including coupled coils formed in the RDL and adjacent BDL for improved Q factor;

[0011] FIG. 2C is a close-up view of another exemplary die interconnect structure with an embedded inductor including coupled coils formed in the RDL and adjacent BDL for improved Q

factor;

[0012] FIG. 3A is top view of an exemplary embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor;

[0013] FIG. 3B is a cross-sectional side view of the embedded inductor along cut line A1 in FIG. 3A illustrating a via which couples the coils of the embedded inductor in FIG. 3A;

[0014] FIG. 3C is top view of another exemplary embedded inductor including coupled coils formed in the RDL and adjacent BDL for improved Q factor;

[0015] FIG. 3D is top view of another exemplary embedded inductor in the shape of a triangle, the exemplary embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor;

[0016] FIG. 3E is top view of another exemplary embedded inductor in the shape of a quadrilateral, the exemplary embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor;

[0017] FIG. 3F is top view of another exemplary embedded inductor in the shape of a hexagon, the exemplary embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor;

[0018] FIG. 3G is top view of another exemplary embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, wherein the coil formed in the RDL is coincident with the coil formed in the BDL;

[0019] FIG. 3H is top view of another exemplary embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, wherein the coil formed in the BDL overlaps the coil formed in the RDL;

[0020] FIG. 4 is a flowchart illustrating an exemplary fabrication process of fabricating a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, including, but not limited to, the exemplary die interconnect structures in FIGS. 2A-2C and embedded inductors in FIGS. 3A-3H;

[0021] FIGS. 5A-5C is a flowchart illustrating another exemplary fabrication process of fabricating a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, including, but not limited to, the exemplary die interconnect structures in FIGS. 2A-2C and embedded inductors in FIGS. 3A-3H;

[0022] FIGS. 6A-6H are exemplary fabrication stages during fabrication of the die interconnect structure according to the fabrication process in FIGS. 6A-6C;

[0023] FIG. 7 is a block diagram of an exemplary processor-based system that can include components deployed in an IC package employing a die(s) coupled to a package substrate, wherein the die(s) can include a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, including, but not limited to, the exemplary die interconnect structures in FIGS. 2A-2C and embedded inductors in FIGS. 3A-3H, and according to the exemplary fabrication processes in FIG. 4 and FIGS. 5A-5C; and

[0024] FIG. 8 is a block diagram of an exemplary wireless communications device that includes radio-frequency (RF) components deployed in an IC package, wherein the IC package includes a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, including, but not limited to, the exemplary die interconnect structures in FIGS. 2A-2C and embedded inductors in FIGS. 3A-3H, and according to the exemplary fabrication processes in FIG. 4 and FIGS. 5A-5C.

#### DETAILED DESCRIPTION

[0025] With reference now to the drawing figures, several exemplary aspects of the present disclosure are described. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects. The term “adjacent” as used herein means spatially next to but not necessarily adjoining something as shown in the Figures unless

specifically stated otherwise.

[0026] Aspects disclosed in the detailed description include a die interconnect structure with embedded inductor(s) including coupled coils formed in a redistribution layer (RDL) and adjacent bump-level distribution layer (BDL) for improved Q factor. Related integrated circuit (IC) packages and fabrication methods are also disclosed. The die interconnect structure formed on a die includes the RDL which includes a metal pad and a first coil. The RDL could be a back-end-of-line (BEOL) interconnect structure of a semiconductor die as an example. The die interconnect structure also includes the BDL which includes a secondary metal pad and a second coil. The BDL is a metallization layer which is as thick or thicker than the RDL. The die interconnect structure also includes a seed layer between the first coil and the second coil and is used in depositing the BDL during fabrication and, in particular, forming the die interconnect and second coil. The BDL may be manufactured at a semiconductor and test (SAT) facility. The die interconnect structure includes an inductor comprising the first coil, the second coil, and a via coupling the first and second coils. Rather than be limited to the number of metallization layers in the die and their smaller thickness than the BDL, the additional second coil is also formed in the same thick BDL containing the die interconnects and is coupled to the first coil formed in the RDL through a via. In this manner, the additional second coil formed in the thick BDL has higher conductance and less resistance than a coil formed in metallization layers within the die and, thus, advantageously improving the Q factor of the inductor.

[0027] In this regard, FIG. 1 is a cross-sectional side view of an exemplary IC package **100** that includes a die interconnect structure with an embedded inductor including coupled coils formed in a redistribution layer (RDL) and adjacent bump-level distribution layer (BDL) for improved Q factor. In this example, the exemplary IC package **100** is a three-dimensional (3D) integrated circuit (IC) (3DIC) package **100** that includes die interconnect structures **102A-102B** with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor. The IC package **100** includes a package substrate **103** and an interposer substrate **104**. The package substrate **103** and the interposer substrate **104** commonly route signals and power and, for convenience, may both be referred to simply as a substrate **106**.

[0028] In this example, the IC package **100** includes first and second dies **108(1)**, **108(2)** that are included in respective first and second die packages **112(1)**, **112(2)** that are stacked on top of each other in the vertical direction (Z-axis direction). The first die package **112(1)** of the IC package **100** includes the first die **108(1)** coupled to the package substrate **103**. In this example, the package substrate **103** includes a first, upper and outer metallization layer **114**. The first, upper and outer metallization layer **114** provides an electrical interface for signal routing to the first die **108(1)**. The first die **108(1)** is coupled to die interconnects **118** (e.g., raised metal bumps, pillars) that are electrically coupled to metal interconnects **120** in the first, upper and outer metallization layer **114**. The first die **108(1)** includes the die interconnect structure **102A** which couples the die interconnects **118** to the circuitry within the first die **108(1)** and includes an embedded inductor with a coil formed in the RDL and coupled to an adjacent coil formed in the BDL. The interconnect structures **102A-102B** will be discussed in more detail in connection with FIGS. 2A-2D. The metal interconnects **120** in the first, upper metallization layer **114** are coupled to metal vias **122** (not visible) in the package substrate **103**, which are coupled to metal interconnects **124** in a second, bottom and outer metallization layer **116**. In this manner, the package substrate **103** provides interconnections between its first and second metallization layers **114** and **116** to provide signal routing to the first die **108(1)**. Both the first and second metallization layers **114** and **116** will contain a die interconnect structures **102A** and **102B**, respectively, and will be discussed in more detail in connection with FIGS. 3A-3D. External interconnects **126** (e.g., ball grid array (BGA) interconnects, a.k.a. bumps) are coupled to the metal interconnects **124** in the second, bottom and outer metallization layer **116** to provide interconnections through the package substrate **103** to the first die **108(1)** through the die interconnects **118**. In this example, a first, active side **128(1)** of the

first die **108(1)** is adjacent to and coupled to the package substrate **103**, and more specifically the first, upper and outer metallization layer **114** of the package substrate **103**.

[0029] In the exemplary IC package **100** in FIG. **1**, an additional optional second die package **112(2)** is provided and coupled to the first die package **112(1)** to support multiple dies. For example, the first die **108(1)** in the first die package **112(1)** may include an application processor, and the second die **108(2)** may be a memory die, such as a dynamic random access memory (DRAM) die that provides memory support for the application processor. In this regard, in this example, the first die package **112(1)** also includes the interposer substrate **104** that is disposed on a package mold **130** encasing the first die **108(1)**, adjacent to a second, inactive side **128(2)** of the first die **108(1)**. The interposer substrate **104** also includes one or more metallization layers **132** that each include metal interconnects **134** to provide interconnections to the second die **108(2)** in the second die package **112(2)**. The second die package **112(2)** is physically and electrically coupled to the first die package **112(1)** by being coupled through external interconnects **136** (e.g., solder bumps, BGA interconnects) to the interposer substrate **104**. The external interconnects **136** are coupled to the metal interconnects **134** in the interposer substrate **104** through metal vias **138** (not visible). The first die package **112(1)** includes vertical interconnects **140** to couple the second die **108(2)** to the external interconnects **126** and to the first die **108(1)** through the package substrate **103**. The second die **108(2)** also includes a die interconnect structure, die interconnect structure **102B** which couples the external interconnects **136** to the circuitry within the second die **108(2)** and includes and includes an embedded inductor with a coil formed in the RDL and coupled to an adjacent coil formed in the BDL. The die interconnect structures **102A-102B** will be discussed in more detail in connection with FIGS. **2A-2C**.

[0030] FIGS. **2A-2C** illustrate three exemplary die interconnect structures with an embedded inductor. FIG. **2A** is a close-up cross-sectional side view of a die shown in FIG. **1**, such as the die **108(1)** or the die **108(2)**, having an exemplary die interconnect structure **200** with an embedded inductor **202** including coupled coils formed in a RDL **204** and an adjacent BDL **206** for improved Q factor. The BDL **206** is a metallization layer that may be manufactured at various widths. Preferably, the width of the BDL **206** in the vertical, Z-axis direction is as wide as the RDL **204** in the vertical, Z-axis direction.

[0031] For this example, the die **108(1)** is shown rotated 180° from the way the die **108(1)** is shown in FIG. **1**. The die **108(1)** includes a first surface **208**. The exemplary die interconnect structure **200** includes the RDL **204** extending in a first, horizontal direction (X-, Y-axes direction) and a second surface **210**. The RDL **204** includes a first coil **212** extending in the first, horizontal direction. The RDL **204** includes metal pad(s) **216** and traces not shown. The BDL **206** also extends in the first, horizontal direction and includes a second coil **214**. The BDL **206** includes secondary metal pad(s) **218** adjacent to the metal pad(s) **216** upon which die interconnects **220A-220C** (e.g., Cu pillars, posts, and the like) are supported. The die interconnect structure **200** includes a dielectric layer **222** (e.g., also known as a passivation layer) extending in the first, horizontal direction and is directly adjacent to the first surface **208** of the die **108(1)** and the first coil **212**. The dielectric layer **222** protects the die **108(1)** before fabricating the secondary metal pad(s) **218** and die interconnects **220A-220C** since the fabrication of the secondary metal pad(s) **218** and die interconnects **220A-220C** may be done at a different fabrication facility than the one that fabricated the die **108(1)**. The dielectric layer **222** also insulates the first coil **212** from the second coil **214**. The height H1 in the vertical direction (Z-axis) of the RDL **204** is approximately 2 μm and the height H2 of the BDL **206** in the vertical direction (Z-axis) can be in the range of 2-10 μm, inclusively.

[0032] The die interconnect structure **200** also includes a seed layer **224** composing of Ti/Cu and extending in the first, horizontal direction adjacent the second surface **210** of the RDL **204** which coincides with the same surface of the metal pad(s) **216** and the same surface of the first coil **212**. The seed layer **224** also is adjacent to the second coil **214** and is utilized to fabricate the secondary metal pad(s) **218** and the second coil **214** as will be discussed in connection with FIGS. **4** and **6A-**

6C. The portion of the seed layer **224** under the secondary metal pad(s) **218** is known as under bump metal. In both respects, the seed layer **224** is between the RDL **204** and the BDL **206** and, thus, between the first coil **212** and the second coil **214**. The embedded inductor **202** includes the first coil **212** and the second coil **214** and a via extending in a second, vertical direction (Z-axis direction) orthogonal to the first direction. The via couples the first coil to the second coil and will be discussed in connection with FIGS. 3A-3B.

[0033] Inductance is proportional to the area enclosed by an inductor's coils times the number of coils. FIGS. 2B and 2C illustrate embedded inductors with additional coils in an exemplary die. FIG. 2B is a close-up view of another exemplary die interconnect structure **226** with an embedded inductor **228** including coupled coils formed in the RDL **204** and the adjacent BDL **206** for improved Q factor. The first coil **212** has a third surface **230** opposite the second surface **210**. The die interconnect structure **226** includes a metallization layer **232** extending in the first direction and adjacent to the third surface **230**. The metallization layer **232** includes traces **234** and a third coil **236** parallel to the first coil **212**.

[0034] FIG. 2C is a close-up view of another exemplary die interconnect structure **238** with an embedded inductor **240** including coupled coils formed in the RDL **204** and the adjacent BDL **206** for improved Q factor. The third coil **236** has a fourth surface **242**. The die interconnect structure **238** includes a metallization layer **244** extending in the first direction and adjacent to the fourth surface **242**. The metallization layer **244** includes traces **246** and a fourth coil **248** parallel to the third coil **236**.

[0035] FIG. 3A is top view of an exemplary embedded inductor **300**, such as the inductors **202**, **228**, and **240** in FIGS. 2A-2C including coupled coils formed in a RDL and adjacent BDL **206** for improved Q factor. The embedded inductor **300** includes a first coil **302** formed in the RDL **204** (dotted perimeter to reflect that the RDL **204** is a lower plane in the vertical Z-direction) and a second coil **304** formed in the BDL. The second coil **304** is parallel to the first coil **302** and is thinner in the X, Y-axes direction than the first coil **302** and, in other words, is said to be coextensive to the first coil **302**. The embedded inductor **300** is in the form of an octagon and the area encompassed by the embedded inductor **300** is approximately **100** square micrometers ( $\mu\text{m}^2$ ). The second coil **304** has a gap **306** where the second coil **304** ends and where the second coil **304** couples to the first coil **302** through a via. The first coil **304** has a gap **307**.

[0036] FIG. 3B is a cross-sectional side view of the embedded inductor **300** along cut line A1 in FIG. 3A illustrating a via **308** which couples the first coil **302** to the second coil **304** in FIG. 3A. The second coil **304** has the gap **306** separating a first end **310** and a second end **312** of the second coil **304**. The first coil **302** has a gap **314** separating a first end **316** and a second end **318** of the first coil **302**. The via **308** is fabricated on a seed layer **320** deposited on a passivation layer **322** and a portion **324** of the first coil **302**. The embedded inductor **300** is provided on a die **326**.

[0037] Embedded inductors which include coupled coils formed in a RDL and adjacent BDL may be formed into various polygons. FIGS. 3C-3F illustrate four exemplary polygons as such an embedded inductor may be fabricated. FIG. 3C is top view of another exemplary embedded inductor **328** including coupled coils formed in a RDL **204** and adjacent BDL **206** for improved Q factor. The embedded inductor **328** includes a first coil **330**, a second coil **332** and a gap **334** between a first end **336** and a second end **338** of the second coil **332**. The first coil **330** includes a gap **339** between two ends of the first coil **330**. The embedded inductor **328** is in the shape of an octagon.

[0038] FIG. 3D is top view of another exemplary embedded inductor **340** in the shape of a triangle, the exemplary embedded inductor **340** including coupled coils formed in a RDL **204** and adjacent BDL **206** for improved Q factor. The embedded inductor **340** includes a first coil **342**, a second coil **344** and a gap **346** between a first end **348** and a second end **350** of the second coil **344**. The first coil **342** includes a gap **351** between two ends of the first coil **342**.

[0039] FIG. 3E is top view of another exemplary embedded inductor **352** in the shape of a

quadrilateral, the exemplary embedded inductor **352** including coupled coils formed in a RDL **204** and adjacent BDL **206** for improved Q factor. The embedded inductor **352** includes a first coil **354**, a second coil **356** and a gap **358** between a first end **360** and a second end **362** of the second coil **356**. The first coil **354** includes a gap **363** between two ends of the first coil **354**.

[0040] FIG. 3F is top view of another exemplary embedded inductor **364** in the shape of a hexagon, the exemplary embedded inductor including coupled coils formed in a RDL **204** and adjacent BDL **206** for improved Q factor. The embedded inductor **364** includes a first coil **366**, a second coil **368** and a gap **370** between a first end **372** and a second end **374** of the second coil **368**. The first coil **366** includes a gap **375** between two ends of the first coil **366**.

[0041] FIG. 3A and FIGS. 3C-3F illustrate that the first coil having a width in the first, horizontal direction (X-, Y-axes direction) is wider than the width of the second coil. Although FIG. 3A and FIGS. 3C-3F illustrate the first coil having a width in the first, horizontal direction (X-, Y-axes direction) that is wider than the width of the second coil, the same polygons may be fabricated where the first coil and the second coil having the same widths such that the perimeters of the first and second coils are the same and are said to be coincident. For example, FIG. 3G is top view of another exemplary embedded inductor **376** including coupled coils formed in a RDL **204** and adjacent BDL **206** for improved Q factor, wherein a first coil **377** formed in the RDL **204** is coincident with a second coil **378** formed in the BDL **206** such that perimeter of the first coil is not shown. As shown in FIG. 3G, the first coil **377** formed in the RDL **204** (not shown in FIG. 3G since its borders coincide with the borders of the coil formed in the BDL **378**) has the same width in the first, horizontal direction (X-, Y-axes direction) as the second coil **378** formed in the BDL **206**. Additionally, the same polygons may be fabricated where the first coil formed in the RDL has a width that is less than the width of the second coil formed in the BDL. For example, FIG. 3H is top view of another exemplary embedded inductor **380** including coupled coils formed in a RDL **204** and adjacent BDL **206** for improved Q factor, wherein the perimeter of the second coil **382** formed in the BDL **206** encompasses the perimeter of the first coil **384** formed in the RDL **204**. The embedded inductor **380** also includes a gap **386** between a first end **388** and a second end **390** of the first coil **384** formed in the RDL **204**. The second coil **382** includes a gap **392** between two ends of the second coil **382**.

[0042] A die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL, including, but not limited to, the embedded inductors **202**, **228**, **240** in FIGS. 2A-2C and the embedded inductors **300**, **328**, **340**, **352**, and **364** in FIGS. 3A-3H in the related IC package **100** in FIG. 1 can be fabricated by different fabrication processes. FIG. 4 is a flowchart illustrating an exemplary fabrication process **400** of fabricating a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, including, but not limited to, the exemplary die interconnect structures in FIGS. 2A-2C and embedded inductors in FIGS. 3A-3H.

[0043] In this regard, a first exemplary step for forming a die interconnect structure formed on a die comprising a first surface **208** in the fabrication process **400** of FIG. 4 can include depositing a RDL **204** extending in a first direction adjacent to the first surface **208** and having a second surface **210** (block **402** in FIG. 4). The step of depositing the RDL **204** can include forming a first coil **212**, **302**, **330**, **342**, **354**, **366** extending in the first direction (block **404** in FIG. 4). The next step in the fabrication process **400** can include depositing a seed layer **224** extending in the first direction adjacent the second surface **210** of the RDL **204** (block **406** in FIG. 4). The next step in the fabrication process **400** can include depositing a BDL **206** extending in the first direction (block **408** in FIG. 4). The step of depositing the BDL **206** can include forming a second coil **214**, **304**, **332**, **344**, **356**, **368**, the seed layer **224** between the first coil **212**, **302**, **330**, **342**, **354**, **366** and the second coil **214**, **304**, **332**, **344**, **356**, **368** (block **410** in FIG. 4). The next step in the fabrication process **400** can include forming an inductor **202**, **228**, **240**, **300**, **328**, **340**, **352**, **364** (block **412** in FIG. 4). The step of forming an inductor can include forming a via **308** extending in a second



direction orthogonal to the first direction, the via **308** coupling the first coil **212**, **302**, **330**, **342**, **354**, **366** to the second coil **214**, **304**, **332**, **344**, **356**, **368** (block **414** in FIG. 4).

[0044] Other fabrication processes can also be employed to fabricate a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, including, but not limited to, the embedded inductors **202**, **228**, **240**, **300**, **328**, **340**, **352**, **364** in FIGS. 2A-2C and 3A-3H. In this regard, FIGS. 5A-5C is a flowchart illustrating another exemplary fabrication process **500** of fabricating a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor including, but not limited to, the embedded inductors **202**, **228**, **240**, **300**, **328**, **340**, **352**, **364** in FIGS. 2A-2C and 3A-3H. FIGS. 6A-6H are exemplary fabrication stages during fabrication of the die interconnect structure according to the fabrication process in FIGS. 5A-5C. The fabrication process **500** as shown in the fabrication stages **600A-600H** in FIGS. 6A-6H are in reference to the cross-sectional side view of the embedded inductor **300** along cut line A1 in FIG. 3A, and thus will be discussed with reference to die interconnect structure **200** in FIG. 2A which is deployed on an existing die such as the die **108(1)** in the related IC package **100** in FIG. 1 and embedded inductor **300** which is a top-view of die interconnect structure **200**.

[0045] In this regard, as shown in fabrication stage **600A** in FIG. 6A, an exemplary step in the fabrication process **500** is depositing a dielectric layer such as a passivation layer **322** (e.g., SiN) over a manufactured die **108(1)** (not shown) including a RDL **204** in which the first coil **212** is formed (block **502** in FIG. 5A). The first coil **212** has a first end **602** and a second end **604**. As shown in fabrication stage **600B** in FIG. 6B, a next step in the fabrication process **500** can include patterning a via opening **606** in the passivation layer **322** utilizing conventional lithography techniques to expose the first coil **212** (block **504** in FIG. 5A). As shown in fabrication stage **600C** in FIG. 6C, a next step in the fabrication process **500** can include depositing a seed layer **224** (e.g., Ti/Cu) over the passivation layer **322** and into the opening to the first coil **212** (block **506** in FIG. 5A). As shown in fabrication stage **600D** in FIG. 6D, a next step in the fabrication process **500** can include patterning a photo resist **608** on the seed layer **224** (block **508** in FIG. 5B). As shown in fabrication stage **600E** in FIG. 6E, a next step in the fabrication process **500** can include depositing a BDL **206** (e.g., Cu plating process) on the exposed seed layer **224** to form a second coil **214** (block **510** in FIG. 5B). As shown in fabrication stage **600F** in FIG. 6F, a next step in the fabrication process **500** can include removing the photo resist **608** (e.g., wet or dry etching process) resulting in a gap **306** (block **512** in FIG. 5B). As shown in fabrication stage **600G** in FIG. 6G, a next step in the fabrication process **500** can include removing the portion of the seed layer **224** in the gap **306** (e.g., wet or dry etch process) (block **514** in FIG. 5C). As shown in fabrication stage **600H** in FIG. 6H, a next step in the fabrication process **500** can optionally include depositing a passivation layer **610** over the die **108(1)** (not shown) (block **516** in FIG. 5C).

[0046] Electronic devices that include an IC package, wherein the IC package includes a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, including, but not limited to, inductors including inductors **202**, **228**, **240** in FIGS. 2A-2C and inductors **300**, **328**, **340**, **352**, **364**, **376**, in FIGS. 3A-3H in the related IC package **100** in FIG. 1, and can be fabricated according to, but not limited to, the exemplary fabrication processes in FIGS. 4 and 5A-5C, and according to any aspects disclosed herein, may be provided in or integrated into any processor-based device. Examples, without limitation, include a set top box, an entertainment unit, a navigation device, a communications device, a fixed location data unit, a mobile location data unit, a global positioning system (GPS) device, a mobile phone, a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a tablet, a phablet, a server, a computer, a portable computer, a mobile computing device, laptop computer, a wearable computing device (e.g., a smart watch, a health or fitness tracker, eyewear, etc.), a desktop computer, a personal digital assistant (PDA), a monitor, a computer monitor, a television, a tuner, a radio, a satellite radio, a music player, a digital music player, a portable music

player, a digital video player, a video player, a digital video disc (DVD) player, a portable digital video player, an automobile, a vehicle component, an avionics system, and a multicopter.

[0047] In this regard, FIG. 7 is a block diagram of an exemplary processor-based system **700** that can include components deployed in an IC package employing a die(s) coupled to a package substrate, wherein the die(s) can include a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, including, but not limited to, the exemplary die interconnect structures in FIGS. 2A-2C and embedded inductors in FIGS. 3A-3H, according to the exemplary fabrication processes in FIG. 4 and FIGS. 5A-5C, and according to any exemplary aspects disclosed herein. In this example, the processor-based system **700** may be formed as an IC package **702** such as the IC package **100** in FIG. 1. The processor-based system **700** includes a central processing unit (CPU) **708** that includes one or more processors **710**, which may also be referred to as CPU cores or processor cores. The CPU **708** may have cache memory **712** coupled to the CPU **708** for rapid access to temporarily stored data. The CPU **708** is coupled to a system bus **714** and can intercouple master and slave devices included in the processor-based system **700**. As is well known, the CPU **708** communicates with these other devices by exchanging address, control, and data information over the system bus **714**. For example, the CPU **708** can communicate bus transaction requests to a memory controller **716**, as an example of a slave device. Although not illustrated in FIG. 7, multiple system buses **714** could be provided, wherein each system bus **714** constitutes a different fabric.

[0048] Other master and slave devices can be connected to the system bus **714**. As illustrated in FIG. 7, these devices can include a memory system **720** that includes the memory controller **716** and a memory array(s) **718**, one or more input devices **722**, one or more output devices **724**, one or more network interface devices **726**, and one or more display controllers **728**, as examples. Each of the memory system(s) **720**, the one or more input devices **722**, the one or more output devices **724**, the one or more network interface devices **726**, and the one or more display controllers **728** can be provided in the same or different electronic devices. The input device(s) **722** can include any type of input device, including, but not limited to, input keys, switches, voice processors, etc. The output device(s) **724** can include any type of output device, including, but not limited to, audio, video, other visual indicators, etc. The network interface device(s) **726** can be any device configured to allow exchange of data to and from a network **730**. The network **730** can be any type of network, including, but not limited to, a wired or wireless network, a private or public network, a local area network (LAN), a wireless local area network (WLAN), a wide area network (WAN), a BLUETOOTH™ network, and the Internet. The network interface device(s) **726** can be configured to support any type of communications protocol desired.

[0049] The CPU **708** may also be configured to access the display controller(s) **728** over the system bus **714** to control information sent to one or more displays **732**. The display controller(s) **728** sends information to the display(s) **732** to be displayed via one or more video processor(s) **734**, which process the information to be displayed into a format suitable for the display(s) **732**. The display controller(s) **728** and video processor(s) **734** can be included as ICs in the same or different electronic devices, and in the same or different electronic devices containing the CPU **708**, as an example. The display(s) **732** can include any type of display, including, but not limited to, a cathode ray tube (CRT), a liquid crystal display (LCD), a plasma display, a light emitting diode (LED) display, etc.

[0050] FIG. 8 is a block diagram of an exemplary wireless communications device that includes radio-frequency (RF) components deployed in an IC package, wherein the IC package, such as the IC package **100**, includes a die interconnect structure with an embedded inductor including coupled coils formed in a RDL and adjacent BDL for improved Q factor, including, but not limited to, the exemplary die interconnect structures in FIGS. 2A-2C and embedded inductors in FIGS. 3A-3H, and according to the exemplary fabrication processes in FIG. 4 and FIGS. 5A-5C, and according to any exemplary aspects disclosed herein. The wireless communications device **800** may include or

be provided in any of the above-referenced devices as examples. As shown in FIG. 8, the wireless communications device **800** includes a transceiver **804** and a data processor **806**. The data processor **806** may include a memory to store data and program codes. The transceiver **804** includes a transmitter **808** and a receiver **810** that support bi-directional communications. In general, the wireless communications device **800** may include any number of transmitters **808** and/or receivers **810** for any number of communication systems and frequency bands. All or a portion of the transceiver **804** may be implemented on one or more analog ICs, RF ICs (RFICs), mixed-signal ICs, etc.

[0051] The transmitter **808** or the receiver **810** may be implemented with a super-heterodyne architecture or a direct-conversion architecture. In the super-heterodyne architecture, a signal is frequency-converted between RF and baseband in multiple stages, for example, from RF to an intermediate frequency (IF) in one stage, and then from IF to baseband in another stage for the receiver **810**. In the direct-conversion architecture, a signal is frequency-converted between RF and baseband in one stage. The super-heterodyne and direct-conversion architectures may use different circuit blocks and/or have different requirements. In the wireless communications device **800** in FIG. 8, the transmitter **808** and the receiver **810** are implemented with the direct-conversion architecture.

[0052] In the transmit path, the data processor **806** processes data to be transmitted and provides I and Q analog output signals to the transmitter **808**. In the exemplary wireless communications device **800**, the data processor **806** includes digital-to-analog converters (DACs) **812(1)**, **812(2)** for converting digital signals generated by the data processor **806** into the I and Q analog output signals (e.g., I and Q output currents) for further processing.

[0053] Within the transmitter **808**, lowpass filters **814(1)**, **814(2)** filter the I and Q analog output signals, respectively, to remove undesired signals caused by the prior digital-to-analog conversion. Amplifiers (AMPs) **816(1)**, **816(2)** amplify the signals from the lowpass filters **814(1)**, **814(2)**, respectively, and provide I and Q baseband signals. An upconverter **818** upconverts the I and Q baseband signals with I and Q transmit (TX) local oscillator (LO) signals through mixers **820(1)**, **820(2)** from a TX LO signal generator **822** to provide an upconverted signal **824**. A filter **826** filters the upconverted signal **824** to remove undesired signals caused by the frequency up-conversion as well as noise in a receive frequency band. A power amplifier (PA) **828** amplifies the upconverted signal **824** from the filter **826** to obtain the desired output power level and provides a transmit RF signal. The transmit RF signal is routed through a duplexer or switch **830** and transmitted via an antenna **832**.

[0054] In the receive path, the antenna **832** receives signals transmitted by base stations and provides a received RF signal, which is routed through the duplexer or switch **830** and provided to a low noise amplifier (LNA) **834**. The duplexer or switch **830** is designed to operate with a specific receive (RX)-to-TX duplexer frequency separation, such that RX signals are isolated from TX signals. The received RF signal is amplified by the LNA **834** and filtered by a filter **836** to obtain a desired RF input signal. Down-conversion mixers **838(1)**, **838(2)** mix the output of the filter **836** with I and Q RX LO signals (i.e., LO\_I and LO\_Q) from an RX LO signal generator **840** to generate I and Q baseband signals. The I and Q baseband signals are amplified by AMPs **842(1)**, **842(2)** and further filtered by lowpass filters **844(1)**, **844(2)** to obtain I and Q analog input signals, which are provided to the data processor **806**. In this example, the data processor **806** includes analog-to-digital converters (ADCs) **846(1)**, **846(2)** for converting the analog input signals into digital signals to be further processed by the data processor **806**.

[0055] In the wireless communications device **800** of FIG. 8, the TX LO signal generator **822** generates the I and Q TX LO signals used for frequency up-conversion, while the RX LO signal generator **840** generates the I and Q RX LO signals used for frequency down-conversion. Each LO signal is a periodic signal with a particular fundamental frequency. A TX phase-locked loop (PLL) circuit **848** receives timing information from the data processor **806** and generates a control signal

used to adjust the frequency and/or phase of the TX LO signals from the TX LO signal generator **822**. Similarly, an RX PLL circuit **850** receives timing information from the data processor **806** and generates a control signal used to adjust the frequency and/or phase of the RX LO signals from the RX LO signal generator **840**.

[0056] Those of skill in the art will further appreciate that the various illustrative logical blocks, modules, circuits, and algorithms described in connection with the aspects disclosed herein may be implemented as electronic hardware, instructions stored in memory or in another computer readable medium wherein any such instructions are executed by a processor or other processing device, or combinations of both. The devices and components described herein may be employed in any circuit, hardware component, integrated circuit (IC), or IC chip, as examples. Memory disclosed herein may be any type and size of memory and may be configured to store any type of information desired. To clearly illustrate this interchangeability, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. How such functionality is implemented depends upon the particular application, design choices, and/or design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

[0057] The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0058] The aspects disclosed herein may be embodied in hardware and in instructions that are stored in hardware, and may reside, for example, in Random Access Memory (RAM), flash memory, Read Only Memory (ROM), Electrically Programmable ROM (EPROM), Electrically Erasable Programmable ROM (EEPROM), registers, a hard disk, a removable disk, a CD-ROM, or any other form of computer readable medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a remote station. In the alternative, the processor and the storage medium may reside as discrete components in a remote station, base station, or server.

[0059] It is also noted that the operational steps described in any of the exemplary aspects herein are described to provide examples and discussion. The operations described may be performed in numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary aspects may be combined. It is to be understood that the operational steps illustrated in the flowchart diagrams may be subject to numerous different modifications as will be readily apparent to one of skill in the art. Those of skill in the art will also understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0060] The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations. Thus, the disclosure is not intended to be limited to the examples and designs described herein, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

[0061] Implementation examples are described in the following numbered clauses:

[0062] 1. A die interconnect structure formed on a die comprising a first surface, the die interconnect structure comprising: [0063] a redistribution layer (RDL) extending in a first direction adjacent to the first surface and having a second surface, the RDL comprising: [0064] a first coil extending in the first direction; [0065] a bump-level distribution layer (BDL) extending in the first direction, the BDL comprising: [0066] a second coil; [0067] a seed layer extending in the first direction adjacent to the second surface of the RDL, the seed layer between the first coil and the second coil; and [0068] an inductor, comprising: [0069] the first coil; [0070] the second coil; and [0071] a via extending in a second direction orthogonal to the first direction, the via coupling the first coil to the second coil.

[0072] 2. The die interconnect structure of clause 1, wherein the first coil and the second coil are parallel, wherein the first coil and the second coil have a same polygon shape.

[0073] 3. The die interconnect structure of clause 1 or 2, wherein the first coil and the second coil are coextensive.

[0074] 4. The die interconnect structure of any of clauses 1-3, wherein the same polygon shape is selected from the group consisting of a triangle, a quadrilateral, a hexagon, and an octagon.

[0075] 5. The die interconnect structure of any of clauses 1-4, wherein the first coil has a third surface opposite the second surface, the inductor further comprising: [0076] a first metallization layer extending in the first direction and adjacent to the third surface, the first metallization layer comprising: [0077] a third coil parallel to the first coil.

[0078] 6. The die interconnect structure of clause 5, wherein the third coil has a fourth surface, the inductor further comprising: [0079] a second metallization layer extending in the first direction and adjacent to the fourth surface, the second metallization layer comprising: [0080] a fourth coil parallel to the third coil.

[0081] 7. The die interconnect structure of any of clauses 1-6, wherein: [0082] the RDL further comprises: [0083] a metal pad; and [0084] the BDL further comprises: [0085] a secondary metal pad, wherein the seed layer is between the secondary metal pad and the metal pad.

[0086] 8. The die interconnect structure of any of clauses 1-7, wherein the second coil encompasses an area of approximately 100 square micrometers ( $\mu\text{m}^2$ ).

[0087] 9. The die interconnect structure of any of clauses 1, 2, and 4-8, wherein the first coil and the second coil are coincident.

[0088] 10. The die interconnect structure of any of clauses 1, 2, and 4-8, wherein the second coil has a second perimeter overlaps a first perimeter of the first coil.

[0089] 11. The die interconnect structure any of clauses 1-10 integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; and a multicopter.

[0090] 12. A method for forming a die interconnect structure formed on a die comprising a first

surface, the method comprising: [0091] depositing a redistribution layer (RDL) extending in a first direction adjacent to the first surface and having a second surface, comprising: [0092] forming a first coil extending in the first direction; [0093] depositing a seed layer extending in the first direction adjacent to the second surface of the RDL; [0094] depositing a bump-level distribution layer (BDL) extending in the first direction, comprising: [0095] forming a second coil, the seed layer between the first coil and the second coil; and [0096] forming an inductor, comprising: [0097] forming a via extending in a second direction orthogonal to the first direction, the via coupling the first coil to the second coil.

[0098] 13. The method of clause 12, wherein the first coil and the second coil are parallel, wherein the first coil and the second coil have a same polygon shape.

[0099] 14. The method of clause 12 or 13, wherein the first coil and the second coil are coextensive.

[0100] 15. The method of any of clauses 12-14, wherein the same polygon shape is selected from the group consisting of a triangle, a quadrilateral, a hexagon, and an octagon.

[0101] 16. The method of any of clauses 12-15, wherein the first coil has a third surface opposite the second surface, the method further comprising: depositing a first metallization layer extending in the first direction and adjacent to the third surface, the first metallization layer comprising: a third coil parallel to the first coil.

[0102] 17. The method of clause 16, wherein the third coil has a fourth surface, the method further comprising: [0103] depositing a second metallization layer extending in the first direction and adjacent to the fourth surface, the second metallization layer comprising: [0104] a fourth coil parallel to the third coil.

[0105] 18. The method of any of clauses 12-17, wherein: [0106] depositing the RDL further comprises: [0107] forming a metal pad; and [0108] depositing the BDL further comprises: [0109] forming a secondary metal pad, wherein the seed layer is between the secondary metal pad and the metal pad.

[0110] 19. The method of any of clauses 12-18, wherein the second coil encompasses an area of approximately 100 square micrometers ( $\mu\text{m}^2$ ).

[0111] 20. The method of any of clauses 12, 13, and 15-19, wherein the second coil overlaps the first coil.

## Claims

1. A die interconnect structure formed on a die comprising a first surface, the die interconnect structure comprising: a redistribution layer (RDL) extending in a first direction adjacent to the first surface and having a second surface, the RDL comprising: a first coil extending in the first direction; a bump-level distribution layer (BDL) extending in the first direction, the BDL comprising: a second coil; a seed layer extending in the first direction adjacent to the second surface of the RDL, the seed layer between the first coil and the second coil; and an inductor, comprising: the first coil; the second coil; and a via extending in a second direction orthogonal to the first direction, the via coupling the first coil to the second coil.

2. The die interconnect structure of claim 1, wherein the first coil and the second coil are parallel, wherein the first coil and the second coil have a same polygon shape.

3. The die interconnect structure of claim 1, wherein the first coil and the second coil are coextensive.

4. The die interconnect structure of claim 2, wherein the same polygon shape is selected from the group consisting of a triangle, a quadrilateral, a hexagon, and an octagon.

5. The die interconnect structure of claim 1, wherein the first coil has a third surface opposite the second surface, the inductor further comprising: a first metallization layer extending in the first direction and adjacent to the third surface, the first metallization layer comprising: a third coil

parallel to the first coil.

**6.** The die interconnect structure of claim 5, wherein the third coil has a fourth surface, the inductor further comprising: a second metallization layer extending in the first direction and adjacent to the fourth surface, the second metallization layer comprising: a fourth coil parallel to the third coil.

**7.** The die interconnect structure of claim 1, wherein: the RDL further comprises: a metal pad; and the BDL further comprises: a secondary metal pad, wherein the seed layer is between the secondary metal pad and the metal pad.

**8.** The die interconnect structure of claim 1, wherein the second coil encompasses an area of approximately 100 square micrometers ( $\mu\text{m}^2$ ).

**9.** The die interconnect structure of claim 1, wherein the first coil and the second coil are coincident.

**10.** The die interconnect structure of claim 1, wherein the second coil has a second perimeter overlaps a first perimeter of the first coil.

**11.** The die interconnect structure claim 1 integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; and a multicopter.

**12.** A method for forming a die interconnect structure formed on a die comprising a first surface, the method comprising: depositing a redistribution layer (RDL) extending in a first direction adjacent to the first surface and having a second surface, comprising: forming a first coil extending in the first direction; depositing a seed layer extending in the first direction adjacent to the second surface of the RDL; depositing a bump-level distribution layer (BDL) extending in the first direction, comprising: forming a second coil, the seed layer between the first coil and the second coil; and forming an inductor, comprising: forming a via extending in a second direction orthogonal to the first direction, the via coupling the first coil to the second coil.

**13.** The method of claim 12, wherein the first coil and the second coil are parallel, wherein the first coil and the second coil have a same polygon shape.

**14.** The method of claim 12, wherein the first coil and the second coil are coextensive.

**15.** The method of claim 13, wherein the same polygon shape is selected from the group consisting of a triangle, a quadrilateral, a hexagon, and an octagon.

**16.** The method of claim 12, wherein the first coil has a third surface opposite the second surface, the method further comprising: depositing a first metallization layer extending in the first direction and adjacent to the third surface, the first metallization layer comprising: a third coil parallel to the first coil.

**17.** The method of claim 16, wherein the third coil has a fourth surface, the method further comprising: depositing a second metallization layer extending in the first direction and adjacent to the fourth surface, the second metallization layer comprising: a fourth coil parallel to the third coil.

**18.** The method of claim 12, wherein: depositing the RDL further comprises: forming a metal pad; and depositing the BDL further comprises: forming a secondary metal pad, wherein the seed layer is between the secondary metal pad and the metal pad.

**19.** The method of claim 12, wherein the second coil encompasses an area of approximately 100 square micrometers ( $\mu\text{m}^2$ ).

**20.** The method of claim 12, wherein the second coil overlaps the first coil.

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